

Figure 1: The data points plotted

```
p =
    2.2212    1.1301

octave:4> polyout(p, "x");

2.2212*x^1 + 1.1301

octave:5> plot(x, y, "x", x, polyval(p, x), "-.");
octave:6> xlabel("Inputs");
octave:7> ylabel("Outputs");
octave:8> legend("Data points", "Linear Fit");
octave:9>
```

`polyfit()` takes three arguments, the 'x' values, the 'y' values, and the order of the polynomial to fit – 1 for linear. It then returns the coefficients of the fitted polynomial *p*. `polyout()` displays the polynomial in a more readable format. So, the fitted equation is $y = 2.2212 * x + 1.1301$. `polyval()` takes the polynomial 'p' and the input values 'x', and calculates the output values 'y', as per the polynomial equation. Then we plot them, along with the earlier data points, on the same plot. Figure 2 shows this, 2.2212 being the slope or gradient of the straight line, and 1.1301 being the intercept. Now, we notice that the straight line does not fit exactly, and there is some deviation from the observed points. `polyfit()` fits the polynomial with the minimum deviation for the order we request, but still there is a deviation, as that is the minimum possible. If you want to get the standard deviation for the particular fit, you just need to issue the following code:

```
octave:9> std_dev = sqrt(mean((y - polyval(p, x)).^2))
std_dev = 0.72637
octave:10>
```

We see that this is a small value compared to our output values. But if we are not fully satisfied with this, we may like

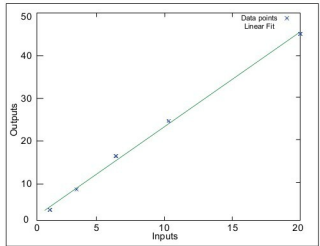


Figure 2: Linear fit of data points

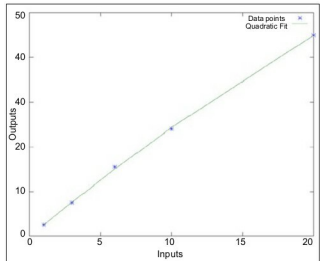


Figure 3: Quadratic fit of data points

to try higher order polynomials, say of order 2, a quadratic equation, and observe the results from it. Here's how one would do this:

```
octave:1> x = [1 3 6 10 20];
octave:2> y = [2.5 7.5 15.5 24 45];
octave:3> p = polyfit(x, y, 2)
p =
   -0.018639    2.623382   -0.051663

octave:4> polyout(p, "x");

-0.018639*x^2 + 2.6234*x^1 - 0.051663

octave:5> plot(x, y, "x", x, polyval(p, x), "-.");
octave:6> xlabel("Inputs");
octave:7> ylabel("Outputs");
octave:8> legend("Data points", "Quadratic Fit");
```